

Practice Question Set

1. What is a data type?
2. Why are data types required in C?
3. Name the three categories of data types in C.
4. What are the basic (primary) data types in C?
5. What are derived data types? Give examples.
6. What are user-defined data types? Give examples.
7. What is a qualifier?
8. What are the different type qualifiers in C?
9. What is the purpose of the short qualifier?
10. What is the purpose of the long qualifier?
11. What is the purpose of the signed qualifier?
12. What is the purpose of the unsigned qualifier?
13. What is the purpose of the const qualifier?
14. Differentiate between float and double.
15. What are the typical memory sizes of char, int, float, and double?

Important Interview/Viva Questions

16. Why are data types important in C?
17. Difference between basic, derived, and user-defined data types.

18. Difference between signed and unsigned integers.

19. Difference between short int and long int.

20. Why is int signed by default?

21. Why are qualifiers used?

22. Can float be used with short, long, or unsigned qualifiers?

23. Why is const considered important?

24. What is long double?

25. Which qualifier preserves a value from being modified?

Answers :

1. What is a data type?

Answer:

A data type is an instruction to the compiler that specifies:

- What kind of data will be stored.
- How much memory should be allocated.

2. Why are data types required in C?

Answer:

Data types help the compiler:

- Identify the type of data.
 - Allocate appropriate memory.
 - Perform operations correctly.
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3. Name the three categories of data types in C.

Answer:

1. Basic (Primary) Data Types
 2. Derived Data Types
 3. User-Defined Data Types
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4. What are the basic data types in C?

Answer:

- char
 - int
 - float
 - double
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5. What are derived data types? Give examples.

Answer:

Derived data types are formed from basic data types.

Examples:

- Arrays
 - Pointers
 - Functions
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6. What are user-defined data types? Give examples.

Answer:

User-defined data types are created by programmers.

Examples:

- struct
- union
- enum

- typedef
-

7. What is a qualifier?

Answer:

A qualifier modifies the properties of a data type.

Examples:

```
short int age;  
long int population;  
unsigned int count;
```

8. What are the different type qualifiers in C?

Answer:

1. short
 2. long
 3. signed
 4. unsigned
 5. const
-

9. What is the purpose of the `short` qualifier?

Answer:

`short` is used to store smaller integer values and consume less memory.

Example:

```
short int age = 20;
```

10. What is the purpose of the `long` qualifier?

Answer:

`long` is used when larger integer values are required.

Example:

```
long int population = 1400000000;
```

11. What is the purpose of the `signed` qualifier?

Answer:

`signed` allows storing both positive and negative values.

Example:

```
signed int temperature = -10;
```

12. What is the purpose of the `unsigned` qualifier?

Answer:

`unsigned` allows storing only positive values.

Example:

```
unsigned int students = 100;
```

Negative values are not allowed.

13. What is the purpose of the `const` qualifier?

Answer:

`const` makes a variable constant, meaning its value cannot be changed.

Example:

```
const float PI = 3.14;  
PI = 5.0; // Error
```

14. Differentiate between `float` and `double`.

Float	Double
Single precision	Double precision
4 bytes	8 bytes
Less accurate	More accurate
Smaller range	Larger range

15. What are the typical memory sizes of common data types?

Data Type Typical Size

char	1 byte
short int	2 bytes
int	4 bytes
float	4 bytes
double	8 bytes

Important Interview Answers

16. Why are data types important in C?

Answer:

Data types specify:

- Type of data to be stored.
 - Amount of memory required.
 - Valid operations on data.
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17. Difference between Basic, Derived, and User-Defined Data Types.

Basic	Derived	User-Defined
Built-in types int, char, float	Built from basic types Arrays, Pointers, Functions	Created by programmer Structure, Union, Enum

18. Difference between Signed and Unsigned Integers.

Signed	Unsigned
Positive and negative values	Positive values only
Default for int	Must be specified
Example: -10, 20	Example: 0, 20

19. Difference between Short Int and Long Int.

Short Int	Long Int
Smaller range	Larger range
Uses less memory	Uses more memory
Used for small values	Used for large values

20. Why is int signed by default?

Answer:

Because an ordinary declaration:

```
int age;
```

is automatically treated as:

```
signed int age;
```

Hence, positive and negative values can be stored.

21. Why are qualifiers used?

Answer:

Qualifiers modify data types to:

- Change range.
 - Change memory usage.
 - Restrict modification.
 - Improve efficiency.
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22. Can float be used with short, long, or unsigned qualifiers?

Answer:

No.

float cannot be combined with:

- short
- unsigned

Only double supports:

long double

23. Why is `const` important?

Answer:

It protects values from accidental modification and improves program reliability.

Example:

```
const float PI = 3.14;
```

24. What is `long double`?

Answer:

`long double` is an extended precision floating-point type that provides greater accuracy than `double`.

Example:

```
long double x;
```

25. Which qualifier prevents modification of a variable?

Answer:

The `const` **qualifier** prevents modification of a variable.

Example:

```
const int MAX = 100;
```